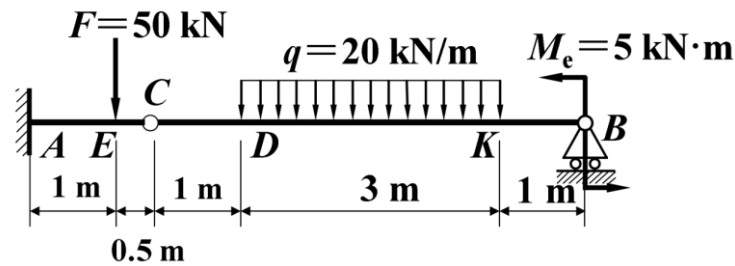
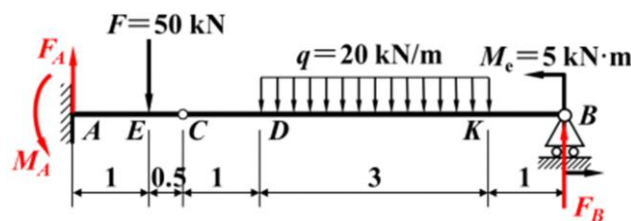


Submission: Please submit the scanned/typed document to Canvas.

1. Draw the shear-force diagram and bending-moment diagram for the following combination beam.



Draw the FBD:



Equilibrium equations:

$$F_A - F - q \times 3 + F_B = 0 \quad (1)$$

Point C can not transfer moment:

✧ Left segment:

$$M_A - F_A(1 + 0.5) + F \times 0.5 = 0 \quad (2)$$

✧ Right segment:

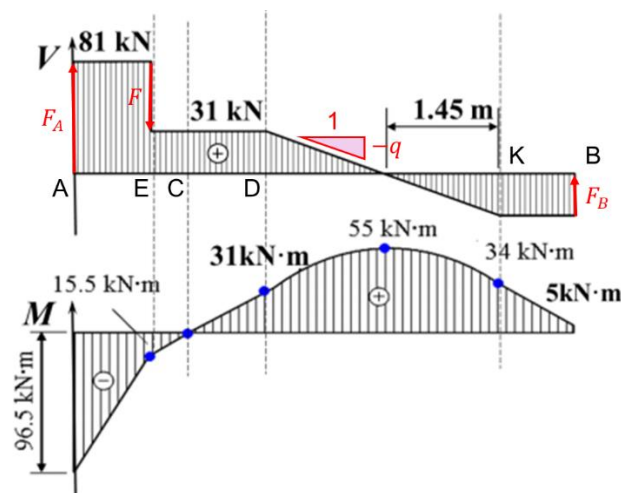
$$-q \times 3 \times \left(1 + \frac{3}{2}\right) + F_B \times (1 + 3 + 1) + M_e = 0 \quad (3)$$

Solve (1), (2), and (3), we have

$$F_A = 81 \text{ kN}, \quad F_B = 29 \text{ kN}, \quad M_A = 96.5 \text{ kN} \cdot \text{m}$$

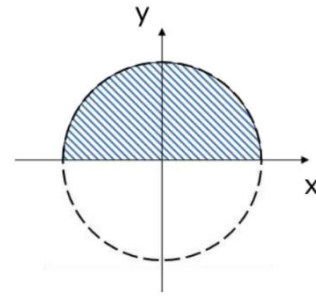
Using jump method, we have the SFD.

Then the BMD can be calculated by integrating the SFD, the starting value at A is $-M_A$



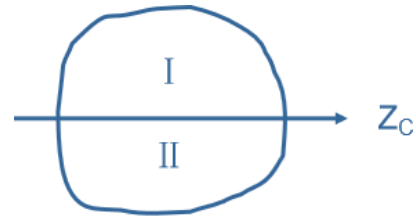
2. For the semicircle shown below, if the circle center is located at the coordinate origin, then (**D**)

- A. $S_x = S_y, I_x \neq I_y$
 B. $S_x = S_y, I_x = I_y$
 C. $S_x \neq S_y, I_x \neq I_y$
 D. $S_x \neq S_y, I_x = I_y$

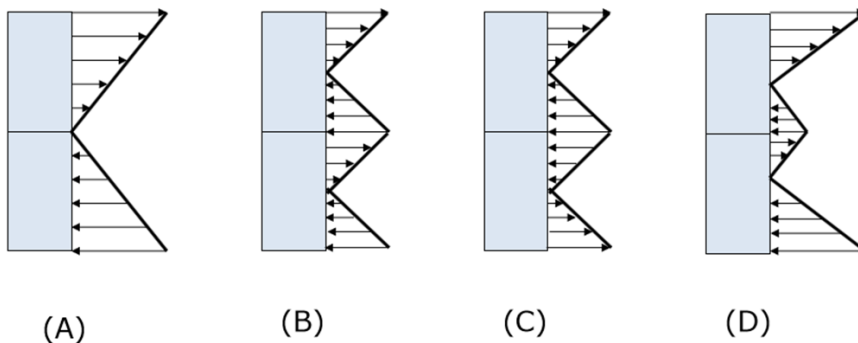
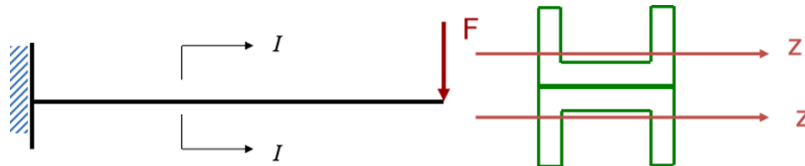


3. For the arbitrary shaped cross-section shown, one centroidal axis z_c (passing through the centroid) divides the section into two parts I and II. Among the following formulas, (**C**) must always hold.

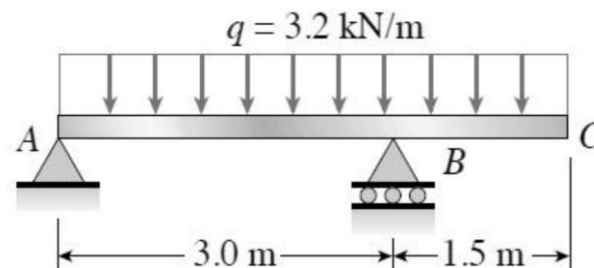
- A. $I_{z_c}^I + I_{z_c}^{II} = 0$
 B. $I_{z_c}^I - I_{z_c}^{II} = 0$
 C. $S_{z_c}^I + S_{z_c}^{II} = 0$
 D. $A^I = A^{II}$



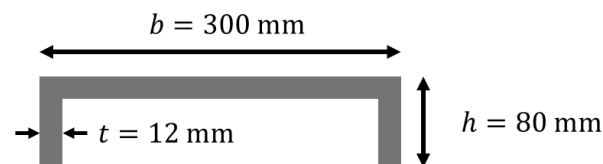
4. A cantilever beam is formed by placing two identical C-channel beams back-to-back (without any connection between them). The overall cross section is shown in the figure below. Under the applied load F , the normal stress distribution on the cross-section is as illustrated in (**D**) (ignore the stress variation in the out-of-plane direction).



5. An overhanging C-channel beam ABC is applied with a uniform load whose intensity is $q = 3.2 \text{ kN/m}$. The cross-section of C-channel shape $b = 300 \text{ mm}$, $h = 80 \text{ mm}$, and the web thickness $t = 12 \text{ mm}$. Determine $\sigma_{\max}$ of both tensile and compressive stress due to the uniform load. (Hint: The difficulty of this problem lies in calculating the moment of inertia of the C-channel beam. Think about the parallel axis theorem.)

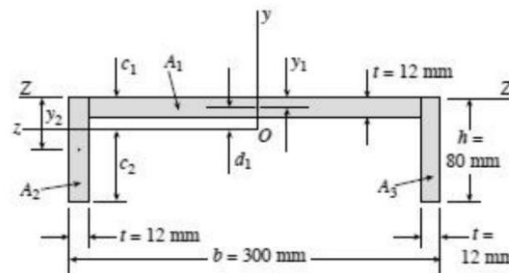


Cross section:



Solution:

First we determine the location of centroid O of the C-channel beam:



Central part:

$$y_1 = 6 \text{ mm}, \quad A_1 = (300 - 2 \times 12) \times 12 \text{ mm}^2 = 3312 \text{ mm}^2$$

Left part:

$$y_2 = 40 \text{ mm}, \quad A_2 = 80 \times 12 \text{ mm}^2 = 960 \text{ mm}^2$$

Right part:

$$y_3 = y_2 = 40 \text{ mm}, \quad A_3 = A_2 = 80 \times 12 \text{ mm}^2 = 960 \text{ mm}^2$$

The distance of centroid to the upper surface of the beam:

$$c_1 = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3} \approx 18.48 \text{ mm}$$

The distance of centroid to the lower surface of the beam:

$$c_2 = h - c_1 \approx 61.52 \text{ mm}$$

Second, we determine the moment of inertia:

$$I_z = I_1 + I_2 + I_3$$

Using parallel axis theorem:

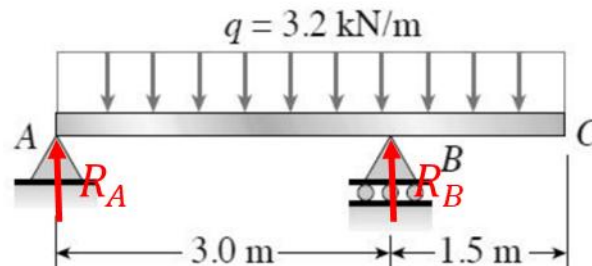
$$I_1 = \frac{(b - 2t)t^3}{12} + \left(c_1 - \frac{t}{2}\right)^2 A_1 = 5.556 \times 10^5 \text{ mm}^4$$

$$I_2 = I_3 = \frac{th^3}{12} + \left(c_2 - \frac{h}{2}\right)^2 A_2 = 9.566 \times 10^5 \text{ mm}^4$$

Therefore,

$$I_z = 2.469 \times 10^6 \text{ mm}^4$$

FBD:



Equilibrium:

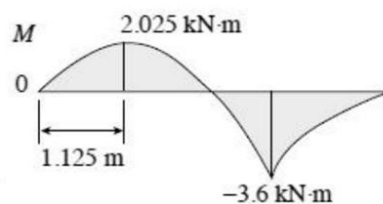
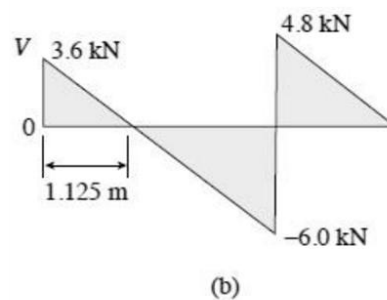
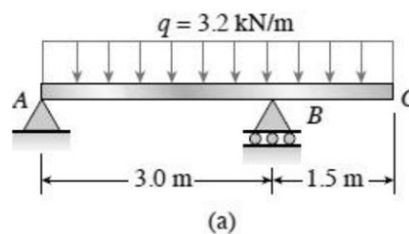
$$R_A + R_B - q(3 + 1.5) = 0$$

$$q(3 + 1.5) \times 0.75 - R_A \times 3 = 0$$

Solved

$$R_A = 3.6 \text{ kN}, \quad R_B = 10.8 \text{ kN}$$

Using Jump method and the differential relation of q , V , and M , we have



The maximum bending moment is $M_{\max}^+ = 2.025 \text{ kN} \cdot \text{m}$, $M_{\max}^- = -3.6 \text{ kN} \cdot \text{m}$

At $M_{\max}^+$:

$$\sigma_t = -\frac{M_{\max}^+(-c_2)}{I_z} = 50.5 \text{ MPa}, \quad \sigma_c = -\frac{M_{\max}^+c_1}{I_z} = -15.2 \text{ MPa}$$

At $M_{\max}^-$:

$$\sigma_t = -\frac{M_{\max}^-c_1}{I_z} = 26.9 \text{ MPa}, \quad \sigma_c = -\frac{M_{\max}^-(-c_2)}{I_z} = -89.8 \text{ MPa}$$

All in all, the maximum tensile stress:

$$\sigma_t = 50.5 \text{ MPa}$$

Maximum compressive stress:

$$\sigma_c = -89.8 \text{ MPa}$$